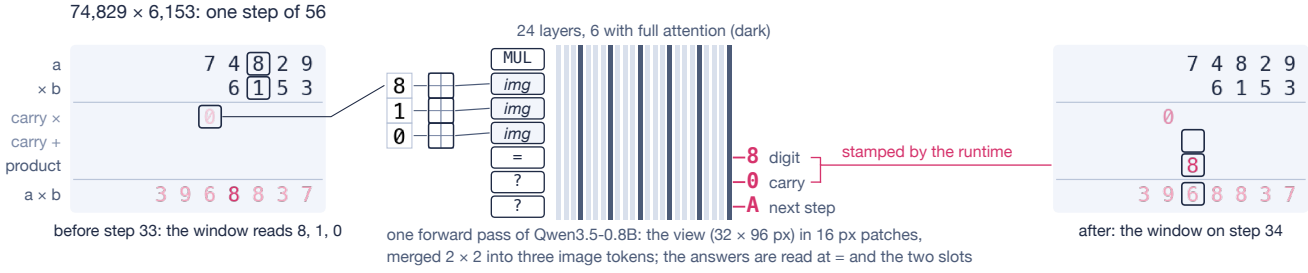
a



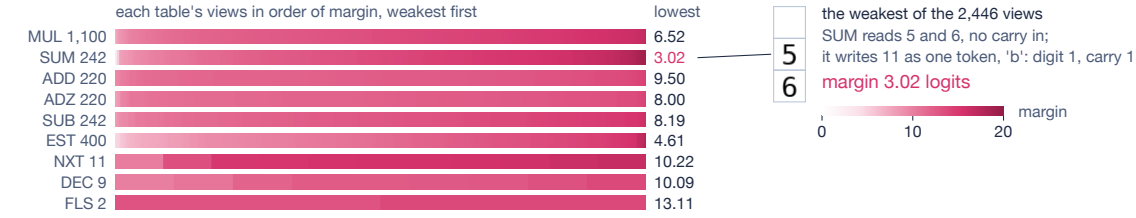


the whole product: 56 steps, one forward pass each, four passes of the multiplier's digits; M multiply A add Z add the last carry F flush N next digit

Between steps the model keeps nothing: the sheet is its only memory.

b

the guarantee: every view the runs can read, checked once



exact on all 2,446 views ⇒ exact on every run of every length (Theorem 1)

2,446 of 2,446 pass on the M4 (WebGPU, fp16); every new GPU checks the weakest tenth again before it runs

c

